

Asymmetric Effects of Sectoral Shifts under Low and High Uncertainty*

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This Version: December 5, 2020

Abstract

This paper studies the asymmetric effects of sectoral shifts on economic performance under low and high uncertainty using U.S. data. A sectoral shift is found to induce more depressed economic activity under high uncertainty relative to under low uncertainty. These effects are statistically different across the two uncertainty regimes and are not driven solely by recessions. A tractable two-sector dynamic stochastic general equilibrium model with sectoral shifts and stochastic volatility is able to qualitatively explain these empirical findings.

Keywords: uncertainty, sectoral shifts, industry stock market dispersion

JEL: E0, E3, G1

*We thank Mario J. Crucini, Carl (Dong) Cheng, Kevin Huang, Nelson Mark, participants at the Midwest Macroeconomic Meetings and Union College, four anonymous referees, and the editor for comments. Financial support from Miami University is greatly appreciated. All errors are our own.

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I INTRODUCTION

The heightened uncertainty experienced during the Great Recession has renewed an interest not only in measuring uncertainty, but also in understanding the macroeconomic implications of varying degrees of uncertainty.¹ This same period was also characterized by considerable structural changes across industries (i.e., sectoral shifts).² Motivated by these recent events, this paper bridges the gap between these historically separately studied topics and examines the asymmetric effects of sectoral shifts on macroeconomic activity under low and high uncertainty. While evidence of the real effects of sectoral shifts has been previously documented in the literature (e.g., [Lilien \(1982\)](#), [Loungani et al. \(1990\)](#), and more recently, [Chodorow-Reich and Wieland \(forthcoming\)](#)), the extent to which these effects depend on the uncertainty regime is unclear.

Our empirical results suggest that sectoral shifts induce more depressed macroeconomic activity under high uncertainty relative to under low uncertainty and these effects are statistically different across the two uncertainty regimes, where the regimes are jointly determined with other parameters. Moreover, although uncertainty is generally found to be counter-cyclical, our results are not driven solely by recessions.³

This paper’s empirical strategy involves employing a threshold regression approach in the spirit of [Gonzalo and Pitarakis \(2002\)](#) and local projections in the spirit of [Jordà \(2005\)](#) using U.S. data. Throughout the empirical analysis, macroeconomic activity is measured by industrial production growth or the unemployment rate change and the main uncertainty measure is macro uncertainty by [Jurado et al. \(2015\)](#) because it is an economy-wide (financial and non-financial) uncertainty measure. Following [Loungani et al. \(1990\)](#) and [Chen et al. \(2011\)](#), we utilize changes

¹See, for example, [Bloom et al. \(2007\)](#), [Bloom \(2009\)](#), [Jurado et al. \(2015\)](#), and [Baker et al. \(2016\)](#).

²For example, [Swanson \(2012\)](#) notes that declines in stock prices for publicly listed home-building firms were followed by layoffs and downsizing that were larger than those of other industries in the economy during the Great Recession. [Jaimovich and Siu \(2018\)](#) document a significant transition from middle-skilled work to either lower-skilled or higher-skilled occupations during the Great Recession. They also attribute the jobless recoveries during the Great Recession to the disappearance of routine occupations. Similarly, [Acemoglu and Autor \(2010\)](#) highlight a significant decline in real wages of low skill workers along with rapid diffusion of technology, both of which are particularly notable during the Great Recession.

³[Schwert \(1990\)](#) and [Jurado et al. \(2015\)](#) provide evidence on the counter-cyclical of uncertainty.

in industry stock market dispersion to proxy for sectoral shifts in the empirical analysis. To construct monthly industry stock market dispersion, we first calculate daily returns for 33 non-overlapping industry sector stock market indices obtained from Datastream Refinitiv. We then calculate the weighted standard deviation across all industry sectors (financial and non-financial), where the weights are the corresponding industry sector number of employee cross-sectional shares. Lastly, we average these daily standard deviation values within each month to arrive at a monthly industry stock market dispersion measure that spans from 1985M1 to 2018M12, which is also the maximal time period of the empirical analysis.

The use of changes in industry stock market dispersion to proxy for shifts in demand from one sector to others has a long intellectual history. Similar to [Loungani et al. \(1990\)](#), our interest in using changes in industry stock market dispersion to capture sectoral shifts can be traced back to [Black \(1987\)](#), who suggests that “when stocks in a given sector go up, more often than not that sector will show a rise in sales, earnings, and outlays for plant and equipment [...] [Therefore,] large moves in opposite directions in different sectors should be followed, normally, by an increase in unemployment.” Intuitively, stock market dispersion across industries is indicative of differentiated shocks to profits of various industries, which facilitates the interpretation of a sectoral shift from one sector to others. In other words, a change in industry stock market dispersion is indirectly capturing the structural shift from the less productive sectors to the more productive sectors. Because these shocks may result in displacements in resources that might not be immediately offset by more productive industries, an increase in this dispersion measure is expected to precede depressed macroeconomic activity. In theory, a change in the industry stock market dispersion measure can capture both the structural change and the level of aggregate demand ([Abraham and Katz, 1986](#)). As noted by [Loungani et al. \(1990\)](#) though, one key advantage of using a change in industry stock market dispersion is that this measure largely captures the change in the structure, which is what we are interested in, and not the

change in the level of aggregate demand.⁴ This approach is consistent with [Brainard and Cutler \(1993\)](#), who also construct a measure of cross-sectional volatility based on market returns and find this measure to largely capture reallocations across industry sectors.

To complement the empirical analysis and provide more structure, we introduce a sector-specific productivity shock (to capture a sectoral shift) and an economy-wide stochastic volatility component (to generate low and high uncertainty) into a tractable two-sector dynamic stochastic general equilibrium (DSGE) model in the spirit of [Moro \(2012\)](#), and study the asymmetric effects of sectoral shifts under low and high uncertainty. The use of stochastic volatility to model uncertainty is motivated, in part, from a recent strand of literature that studies the real implications of uncertainty ([Fernández-Villaverde et al., 2011, 2015](#)). Our choice of a sector-specific productivity shock to model a sectoral shift in the DSGE model is motivated by our empirical analysis as well as the sectoral shift literature ([Abraham and Katz, 1986](#); [Loungani et al., 1990](#)). In the model, a negative productivity shock in one sector and not in the other sector captures the shift from the relatively less productive sector to the relatively more productive sector and is not necessarily reflective of the change in the level of aggregate demand, as is the case with our empirical measure of a sectoral shift. When a reduction in productivity in one sector is not perfectly offset by a corresponding relative increase in productivity in the other sector, aggregate output decreases.

Parameterized to capture salient features of the U.S. manufacturing and service sectors as in [Moro \(2012\)](#), the model predicts that a sectoral shift induces lower output growth during periods of high uncertainty relative to during periods of low uncertainty, which is qualitatively consistent with the empirical results. The model also predicts asymmetric output responses across the two uncertainty regimes regardless of whether the sectoral shift is initiated when output is *ex-ante* low (i.e., in a recessionary state) or *ex-ante* high (i.e., in an expansionary state). This prediction aligns qualitatively with the empirical result that the asymmetric effects of sectoral shifts under

⁴The literature finds changes in industry stock market dispersion to largely capture sectoral reallocation since industry stock market dispersion's correlation with unemployment is largely not attributable to aggregate demand movements ([Loungani et al., 1990](#); [Neumann and Topel, 1991](#)). We also note that our measure of industry stock market dispersion is moderately positively correlated with the cross-sectional standard deviation of firm-level revenue divided by the number of employees ([Gabaix, 2011](#)).

low and high uncertainty are not driven solely by recessions.

More generally, this paper is related to two main strands of literature. The first strand documents the impact of uncertainty on the economy both empirically and theoretically (Schwert, 1990; Bloom et al., 2007; Bloom, 2009; Fernández-Villaverde et al., 2011, 2015; Jurado et al., 2015; Baker et al., 2016; Bloom et al., 2018; Kalay et al., 2018). The second strand documents the impact of sectoral shifts on aggregate demand (Lilien, 1982; Grossman et al., 1983; Abraham and Katz, 1986; Loungani et al., 1990; Neumann and Topel, 1991; Brainard and Cutler, 1993; Kalay et al., 2018; Chodorow-Reich and Wieland, forthcoming). These two strands of literature are for the most part, distinct and separate from one another. Kalay et al. (2018) is one exception that lies at the intersection of the two and is most closely related to our paper. Kalay et al. (2018) employ an OLS regression approach with a firm-level earnings dispersion and uncertainty dummy interaction term to document a statistically significant relationship between this interaction term and macroeconomic activity. The uncertainty measures Kalay et al. (2018) consider include an aggregate earnings uncertainty measure, the Baker et al. (2016) economic policy uncertainty (EPU) index, and the Chicago Board Options Exchange (CBOE) S&P 500 Volatility Index (VIX). While novel in their own rights, Kalay et al. (2018) do not show whether the effects on macroeconomic activity are statistically different across the low and high uncertainty regimes and do not estimate the implied uncertainty threshold jointly with the other parameters of interest.

The key contribution of our paper, therefore, is to document the *asymmetric effects* of sectoral shifts, as proxied by increases in industry stock market dispersion, on macroeconomic activity under low and high uncertainty. Unlike Kalay et al. (2018), our preferred empirical measure of uncertainty is Jurado et al. (2015) macro uncertainty because it is a relatively comprehensive measure of economy-wide uncertainty. The Jurado et al. (2015) macro uncertainty measure is based on an econometric analysis of 132 financial and real (non-financial) time-series data and is the equally-weighted average of individual uncertainty measures of each financial and real time-series. We consider macro uncertainty measures with forecast horizons of one-month (H-1), three-months (H-3), and twelve-months (H-12). As noted in Jurado et al. (2015), these macro

uncertainty measures do not vary over time when there is no change in the underlying economy-wide fundamentals, as other uncertainty measures do. To align our paper with [Kalay et al. \(2018\)](#) though, we also consider the commonly used more market-specific uncertainty measures of the [Baker et al. \(2016\)](#) economic policy uncertainty index (EPU) and the Chicago Board Options Exchange (CBOE) S&P 100 Volatility Index (VXO), and find no to weak asymmetric effects.⁵ Additionally, in contrast to [Kalay et al. \(2018\)](#), we provide more structure in support of the documented empirical asymmetric effects of sectoral shifts via a tractable dynamic two-sector model. Our result that the asymmetric effects of sectoral shifts under low and high uncertainty are not driven solely by recessions also complements [Chodorow-Reich and Wieland \(forthcoming\)](#), who find amplified effects of a sectoral reallocation during a national recession using U.S. data from 1980 to 2014.

The remainder of the paper proceeds as follows. Section II describes the data. Section III presents the empirical methodology and main results. Section IV outlines the two-sector dynamic stochastic general equilibrium model and discusses the implications of the model. Section V concludes.

II DATA

The data used in the empirical analysis are monthly U.S. data and have a maximal span from 1985M1 to 2018M12. The raw data are either daily or monthly and consist of industrial production, unemployment rate, the federal funds rate, macro uncertainty, economic policy uncertainty, the Chicago Board Options Exchange (CBOE) S&P 100 Volatility Index (VXO), industry sector stock market indices, and industry sector numbers of employees. Industrial production, the unemployment rate, the federal funds rate, and the Chicago Board Options Exchange (CBOE) S&P 100 Volatility Index (VXO) are taken from the FRED database at the Federal Reserve

⁵The [Baker et al. \(2016\)](#) economic policy uncertainty index captures uncertainty regarding economic policy and the Chicago Board Options Exchange (CBOE) S&P 100 Volatility Index (VXO) captures uncertainty arising in the stock market. The cross-sectional dispersion of GDP forecasts also has been used as a proxy for uncertainty in the literature, however, this measure is not available at the monthly frequency so we abstract from using this measure in the paper. We also note that the cross-sectional dispersion of GDP forecasts may be reflective of differences in opinion or biases rather than overall economic uncertainty.

Bank of St. Louis. Data for macro (H-1, H-3, and H-12) uncertainty are from [Jurado et al. \(2015\)](#).⁶ Here, H-1, H-3, and H-12 denote the forecast horizons (in months) from which the uncertainty indices are constructed. Economic policy uncertainty (EPU) is from [Baker et al. \(2016\)](#).⁷ The industry sector stock market indices and the industry sector number of employees are from Datastream.⁸

For the empirical analysis, we construct monthly industrial production growth by first seasonally adjusting the industrial production series (moving average of the current and previous twelve months of the variable) and then taking the log first difference of the seasonally adjusted industrial production series. We construct the monthly change in the unemployment rate by first seasonally adjusting the unemployment rate series (moving average of the current and previous twelve months of the variable) and then taking the first difference of the seasonally adjusted unemployment rate series.⁹

[Table 1 Here]

Monthly industry stock market dispersion is constructed from the daily industry sector stock market indices for the U.S. (US-DS), which consists of 33 non-overlapping industry sector stock market indices. These indices are constructed from all publicly traded firms within each industry sector. The names of the 33 industry sectors (financial and non-financial) and the corresponding industry sector i 's time-series average of the cross-sectional share of employees ($\bar{c}_i$) are listed in Table 1.

To construct monthly industry stock market dispersion, we first construct daily returns ($R_{i,t}$) for each industry sector i in day t from the daily industry sector stock market indices ($D_{i,t}$),¹⁰

$$R_{i,t} = \frac{D_{i,t} - D_{i,t-1}}{D_{i,t-1}}. \quad (1)$$

⁶Data for [Jurado et al. \(2015\)](#) macro uncertainty were downloaded from www.sydneyludvigson.com/data-and-appendixes.

⁷[Baker et al. \(2016\)](#) economic policy uncertainty was downloaded from www.policyuncertainty.com.

⁸Additional background information on the indices can be obtained directly from Thompson Reuters or at www.datastream.jp/wp/wp-content/uploads/2017/02/DatastreamGlobalEquityIndicesUGissue05.pdf.

⁹Figure B.1 in Appendix B plots industrial production growth, the unemployment rate change, and the federal funds rate.

¹⁰The daily industry sector stock market indices are normalized so that the values on January 2, 2012 are 100. We use January 2, 2012 because the stock market is closed on the first day of the new year.

We next construct daily industry stock market dispersion σ_t^R by taking the standard deviation across all industry sectors of the daily returns $R_{i,t}$, which is weighted by the corresponding industry sector number of employee cross-sectional share $c_{i,t}$. The use of subscript t for $c_{i,t}$ here is intentional: the number of employees used in the construction of our measure of stock market dispersion is allowed to vary over time. This is to reflect the ever-changing nature of the U.S. economy. To obtain monthly industry stock market dispersion, we average the daily values of σ_t^R within each month. This construction of industry stock market dispersion follows [Loungani et al. \(1990\)](#). Also, following [Loungani et al. \(1990\)](#), a change in this measure represents a sectoral shift in the empirical analysis.

[Figure 1 Here]

Figure 1 plots our constructed monthly industry stock market dispersion series over time along with the uncertainty measures of [Jurado et al. \(2015\)](#) macro (H-1, H-3, and H-12) uncertainty, [Baker et al. \(2016\)](#) economic policy uncertainty, and the Chicago Board Options Exchange (CBOE) S&P 100 Volatility Index (VXO). The grey bars in Figure 1 represent NBER recession dates. Going forward, we will refer to the [Jurado et al. \(2015\)](#) macro uncertainty measure as macro uncertainty, the [Baker et al. \(2016\)](#) economic policy uncertainty index as EPU, and the Chicago Board Options Exchange (CBOE) S&P 100 Volatility Index as VXO. Table 2 provides summary statistics for the variables used in the analysis.

[Table 2 Here]

We note that the [Jurado et al. \(2015\)](#) macro (H-1, H-3, and H-12) uncertainty measures do not reflect the same information as industry stock market dispersion and also are not highly correlated with one another. More specifically, the correlations between the H-1, H-3, and H-12 uncertainty measures and industry stock market dispersion are 0.316, 0.323, and 0.323, respectively.

III EMPIRICAL METHODOLOGY AND RESULTS

This section outlines the empirical methodology we employ to study the asymmetric effects of sectoral shifts, as proxied by increases in industry stock market dispersion, on macroeconomic activity under different uncertainty regimes, as well as presents our main empirical results. In Section III.A, we consider a threshold regression approach. The advantage of this approach is that the threshold parameter (i.e., which determines the low and high uncertainty regimes) is estimated jointly with the other parameters. In Section III.B, we use local projections in the spirit of [Jordà \(2005\)](#) to produce impulse responses under alternative uncertainty regimes.

III.A Threshold Regression

To study the asymmetric effects of industry stock market dispersion on macroeconomic aggregates under low and high uncertainty, we consider the following threshold regression in the spirit of [Gonzalo and Pitarakis \(2002\)](#),

$$\begin{aligned}
 y_t = & \sum_{j=1}^k \beta_j y_{t-j} + \sum_{j=1}^k \psi_j \Delta i_{t-j} + \xi D_t + \sum_{j=1}^k \delta_j z_{t-j} I(-\infty < \omega_t \leq \gamma) + \alpha_1 I(-\infty < \omega_t \leq \gamma) + \dots \\
 & + \sum_{j=1}^k \theta_j z_{t-j} I(\gamma < \omega_t < \infty) + \alpha_2 I(\gamma < \omega_t < \infty) + \epsilon_t.
 \end{aligned} \tag{2}$$

Here, y_t is a measure of macroeconomic activity (industrial production growth or the unemployment rate change) at time t , i_t is the nominal interest rate at time t , D_t is an NBER recession dummy variable at time t , z_t is industry stock market dispersion at time t , $I(\cdot)$ is the uncertainty regime indicator function, and ω_t is the threshold variable, which is an uncertainty measure at time t .

Since our interest is in the real effects of sectoral shifts, our identification strategy hinges on the use of changes in industry stock market dispersion (z_t), which captures shifts across sectors, instead of changes in the level of aggregate demand, to capture sectoral shifts. This strategy is motivated by a large and growing related literature on sectoral shifts ([Loungani et al., 1990](#); [Brainard, 1992](#); [Vu and Wu, 2020](#)).

The benchmark threshold variable in our analysis is the macro (H-1, H-3, and H-12) uncertainty measure from [Jurado et al. \(2015\)](#) because it is a relatively comprehensive measure of general uncertainty in the economy, whereas other measures of uncertainty tend to change over time even though there are no changes in the underlying economy-wide fundamentals (as documented by [Jurado et al. \(2015\)](#)). Nevertheless, we also consider two more market-specific uncertainty measures: (1) economic policy uncertainty (EPU) from [Baker et al. \(2016\)](#) and (2) the Chicago Board Options Exchange (CBOE) S&P 100 Volatility Index (VXO), since they are uncertainty measures that are widely used in the literature.¹¹

In equation (2), γ is the threshold parameter that is estimated jointly with the other parameters.¹² Since inference on γ is complicated due to its nonstandard asymptotic distribution ([Hansen, 1997, 2000](#)), we do not report its standard error. Two additional parameters of interest are δ_j and θ_j , which are the parameters on the *lags* of industry stock market dispersion under the two uncertainty regimes. We also note that in equation (2), we control for the lags of the dependent variable, the lags of the change in the nominal interest rate, and U.S. recessions. We select a lag length of $k = 2$ according to the Schwarz information criterion.¹³ The maximal time period used in our empirical analysis is 1985M1 to 2018M12.

We use conditional least squares to estimate the parameters of equation (2). To obtain an estimate for the threshold parameter (γ), we minimize the least squares of equation (2) using all observations between the 10th and 90th percentiles of the threshold variable (ω_t).¹⁴

¹¹We consider the VXO series rather than the VIX series because it has a longer time-series. The two series are highly correlated.

¹²Although our analysis focuses on two regimes (low and high uncertainty) across all specifications, we also tested for the optimal number of thresholds (assuming a maximum of three thresholds) using macro (H-1, H-3, and H-12) uncertainty as the uncertainty measure and $k = 2$ and found the optimal number of thresholds to be one (i.e., two uncertainty regimes).

¹³We tested for the optimal lag length using the Schwarz information criterion (SBIC) with a maximal lag length of twelve. When using macro (H-1, H-3, and H-12) uncertainty as the uncertainty threshold, the optimal lag length is two ($k = 2$).

¹⁴The estimator for the threshold is $\hat{\gamma} = \arg \min_{\gamma} \mathbf{G}(\gamma)$, where,

$$\mathbf{G}(\gamma) = \sum_{t=1}^T \left[y_t - \left(\sum_{j=1}^k \beta_j y_{t-j} + \sum_{j=1}^k \psi_j \Delta i_{t-j} + \xi D_t + \sum_{j=1}^k \delta_j z_{t-j} I(-\infty < \omega_t \leq \gamma) + \alpha_1 I(-\infty < \omega_t \leq \gamma) + \dots + \sum_{j=1}^k \theta_j z_{t-j} I(\gamma < \omega_t < \infty) + \alpha_2 I(\gamma < \omega_t < \infty) \right) \right]^2.$$

[Table 3 Here]

Table 3 presents the threshold regression results where the dependent variable is industrial production growth and the threshold variables are macro (H-1, H-3, and H-12) uncertainty. Specifications (1)–(3) do not include an NBER recession dummy variable in the regression, specifications (4)–(6) include an NBER recession dummy variable in the regression, and specifications (7)–(9) include an NBER Great Recession (December 2007 – June 2009) dummy variable in the regression. The results in Table 3 suggest that increases in industry stock market dispersion are significantly and negatively associated with industrial production growth under the high macro (H-1, H-3, and H-12) uncertainty regime and are weakly associated with industrial production growth under the low macro (H-1, H-3, and H-12) uncertainty regime. In addition, the χ^2 statistics reported in Table 3 suggest that the cumulative impacts of industry stock market dispersion on industrial production growth are statistically different across the low and high macro (H-1, H-3, and H-12) uncertainty regimes at the one-percent significance level. Overall, these results suggest that a sectoral shift, as proxied by an increase in industry stock market dispersion, is associated with a statistically different and larger reduction in industrial production growth under high macro (H-1, H-3, and H-12) uncertainty relative to under low macro (H-1, H-3, and H-12) uncertainty and these results are not driven solely by recessions or the Great Recession, in particular.

[Table 4 Here]

Next, in Table 4, we conduct a similar analysis as in Table 3, but the threshold variable is either EPU or VXO, which are more market-specific uncertainty measures, instead of macro uncertainty. These results are weaker than the macro uncertainty threshold regression results. This may be driven in part by the more market-specific nature of EPU and VXO, whereas macro uncertainty is a more economy-wide uncertainty measure. A more market-specific uncertainty measure may fluctuate when there are no changes in the underlying economy-wide fundamentals and thus may not be reflective of the true aggregate uncertainty state of the economy. For

example, when the aggregate economy is in a low uncertainty state, there may be periods where the more market-specific uncertainty measure also indicates a low uncertainty state and others where it indicates a high uncertainty state and thus, the effects are less distinguishable across the low and high more market-specific uncertainty regimes. We investigate this further in the next subsection.

[Table 5 Here]

We also consider the unemployment rate change as an alternative measure of macroeconomic activity. Table 5 presents the threshold regression results where the threshold variable is macro (H-1, H-3, and H-12) uncertainty. Specifications (1)–(3) do not include an NBER recession dummy variable in the regression, specifications (4)–(6) include an NBER recession dummy variable in the regression, and specifications (7)–(9) include an NBER Great Recession dummy variable in the regression. In Table 5, we also find statistically significant asymmetric effects of industry stock market dispersion on the unemployment rate change under low and high macro (H-1, H-3, and H-12) uncertainty at the one-percent significance level (except in specifications (6) and (9); five-percent significance level) and these results are not driven solely by recessions. Under high macro (H-1, H-3, and H-12) uncertainty, a sectoral shift, as proxied by an increase in industry stock market dispersion, is associated with a statistically significant increase in the unemployment rate change (except in specification (6)).

[Table 6 Here]

Table 6 conducts a similar analysis as in Table 5, but the threshold variable is either EPU or VXO instead of macro uncertainty. Here, we find weaker or no asymmetric effects of sectoral shifts on the unemployment rate change. As discussed above, this could be a result of the more market-specific nature of EPU and VXO.

In sum, the threshold regression results suggest that sectoral shifts, as measured by increases in industry stock market dispersion, under low and high macro (H-1, H-3, and H-12) uncertainty have asymmetric effects on macroeconomic activity (i.e., industrial production growth and the

unemployment rate change). More specifically, sectoral shifts are associated with more depressed economic activity under high macro uncertainty relative to under low macro uncertainty. These effects are statistically different from one another and are not driven solely by recessions.¹⁵ The asymmetric effects are less clear when the threshold variable is either EPU or VXO.¹⁶

III.B Local Projections

In this section, we use an alternative approach to study the asymmetric effects of industry stock market dispersion on macroeconomic aggregates under the low and high uncertainty regimes. Here, we employ local projections in the spirit of [Jordà \(2005\)](#) to produce impulse responses of industrial production growth or the unemployment rate change to a shock to industry stock market dispersion during periods of low and high uncertainty. In other words, this approach allows us to understand how a change in industry stock market dispersion impacts *future* macroeconomic activity over a longer period of time relative to the threshold regression approach. We also note that when there is a change in the level of aggregate demand, it will take time for the resources to be reallocated across sectors. This could be due to the time-to-build effect ([Kydland and Prescott, 1982](#)) or other market frictions ([Mortensen, 2011](#)). On the other hand, reallocation of resources across sectors will be reflected more quickly in the unemployment statistics, for example.

One advantage of this local projections approach is that it is less prone to misspecification, as documented by [Jordà \(2005\)](#) and [Plagborg-Møller and Wolf \(forthcoming\)](#). The local projections are a sequence of regressions estimated separately for horizons $h = 0, \dots, 12$ and also separately during periods of low and high uncertainty.

¹⁵For robustness to the results presented in this section, we extend the recession dummies by three months. These results are reported in Tables A.1, A.2, A.3, and A.4 in Appendix A. The results continue to suggest asymmetric effects of changes in industry stock market dispersion on macroeconomic activity under low and high macro (H-1, H-3, and H-12) uncertainty.

¹⁶Also for robustness, Tables A.5, A.6, A.7, and A.8 in Appendix A present the threshold regression results where the change in the nominal interest rate is not included as a control.

Regime 1: Periods of Low Uncertainty ($-\infty < \omega_t \leq \hat{\gamma}$)

$$y_{t+h} = \alpha_h + \delta_h z_t + \sum_{j=1}^k \beta_{h,j} y_{t-j} + \sum_{j=0}^k \psi_{h,j} \Delta i_{t-j} + \xi_h D_t + \epsilon_{t+h}. \quad (3)$$

Regime 2: Periods of High Uncertainty ($\hat{\gamma} < \omega_t < \infty$)

$$y_{t+h} = \alpha_h + \theta_h z_t + \sum_{j=1}^k \beta_{h,j} y_{t-j} + \sum_{j=0}^k \psi_{h,j} \Delta i_{t-j} + \xi_h D_t + \epsilon_{t+h}. \quad (4)$$

Here, y_t is a measure of macroeconomic activity (industrial production growth or the unemployment rate change) at time t , i_t is the nominal interest rate at time t , D_t is an NBER recession dummy variable at time t , z_t is industry stock market dispersion at time t , ω_t is an uncertainty measure (macro uncertainty, EPU, or VXO) at time t , and $\hat{\gamma}$ is the uncertainty threshold parameter estimated from the corresponding threshold regression in Section III.A.¹⁷ The maximal time period is 1985M1 to 2018M12 and $k = 2$, also following from Section III.A.

[Figure 2 Here]

Figure 2 displays plots of δ_h (low uncertainty regime; dashed line) and θ_h (high uncertainty regime; solid line), which represent the local projections of industry stock market dispersion on industrial production growth with 90% confidence bands (shaded area). Following [Jordà \(2005\)](#), we interpret these projections as the impulse responses of industrial production growth to a one-unit increase in industry stock market dispersion during periods of low and high uncertainty. The impulse responses with the macro (H-1, H-3, and H-12) uncertainty thresholds and without an NBER recession dummy variable in the regression are in the left column of Figure 2 and the corresponding responses with an NBER recession dummy variable in the regression are in the right column of Figure 2. The horizontal axes measure the months following the industry stock market dispersion shock and the vertical axes measure the response of industrial production growth in percent.

¹⁷Note that in the local projections analysis, the industry stock market variable (z_t) is scaled by 1,000 for presentation purposes.

One key insight from Figure 2 is that increases in industry stock market dispersion during periods of high macro (H-1, H-3, and H-12) uncertainty induce negative responses of industrial production growth and the responses up to approximately seven months are statistically different from the responses up to approximately seven months during periods of low macro (H-1, H-3, and H-12) uncertainty. These contrasting impulse responses under the low and high macro uncertainty regimes are not driven solely by recessions and are similar in spirit to the threshold regression results in Section III.A.

[Figure 3 Here]

In Figure 3, we repeat the exercise as in Figure 2, but with EPU or VXO as the uncertainty threshold variable. Overall, we find weak or no statistically significant asymmetric effects of industry stock market dispersion on industrial production growth during periods of low and high uncertainty when EPU and VXO are used as the uncertainty threshold variables.

[Figure 4 Here]

The next set of results consider the unemployment rate change as an alternative measure of macroeconomic activity and are displayed in Figure 4. The impulse responses with the macro (H-1, H-3, and H-12) uncertainty thresholds and without an NBER recession dummy variable in the regression are in the left column of Figure 4 and the impulse responses with an NBER recession dummy variable in the regression are in the right column of Figure 4. The horizontal axes measure the time in months after the industry stock market dispersion shock and the vertical axes measure the response of the unemployment rate change in percentage points.

The impulse responses presented in Figure 4 suggest that a positive shock to industry stock market dispersion during periods of high macro (H-1, H-3, and H-12) uncertainty results in an increase in the unemployment rate change, whereas during periods of low macro (H-1, H-3, and H-12) uncertainty, there is a negligible effect on the unemployment rate change. In particular, the impulse responses up to about seven months under the low macro (H-1, H-3, and H-12) uncertainty regime are generally statistically different from the impulse responses up to about

seven months under the high macro (H-1, H-3, and H-12) uncertainty regime. Moreover, these asymmetric effects are not driven solely by recessions. We do not find asymmetric effects of industry stock market dispersion on the unemployment rate change under low and high EPU and VXO. These results are reported in Figure 5.

[Figure 5 Here]

Under both the local projections and the threshold regression analyses, the effects of sectoral shifts are documented to be consistently asymmetric under the low and high uncertainty regimes only when macro (H-1, H-3, and H-12) uncertainty is used as the uncertainty threshold variable. To understand the source of this result, we note that the macro (H-1, H-3, and H-12) uncertainty measures are only weakly correlated with EPU and VXO and that there are a non-trivial number of periods in our sample that the former measure and the latter two measures do not identify the same low and high uncertainty regimes.¹⁸ Since EPU and VXO are more market-specific uncertainty measures, these series can fluctuate even when there are no changes in economy-wide fundamentals, which macro uncertainty is intended to capture, and as a result, the uncertainty regimes identified under these two more market-specific uncertainty measures do not always coincide with the macro uncertainty measure. This observation aligns with [Jurado et al. \(2015\)](#), who find VXO, for example, to fluctuate even though there are no changes in the underlying fundamentals.

We next provide more structure for the empirical analysis and study the asymmetric effects of sectoral shifts on output under low and high uncertainty using a tractable two-sector DSGE model in Section IV. To illustrate the model counterpart to the empirical exercise while also trying to keep the model relatively simple and straightforward, we consider a two-sector model rather than a model with more than two sectors. Also, since empirically, a change in industry stock market dispersion can be interpreted as capturing the structural shift from the relatively less productive sectors to the relatively more productive sectors, in the DSGE model, for simplicity, a sectoral shift is modeled via a negative productivity shock in one sector and not in the other sector.

¹⁸See Figure B.2 and Figure B.3 in Appendix B.

IV MODEL

In this section, we introduce sectoral shifts and stochastic volatility into a stylized two-sector dynamic stochastic general equilibrium model in the spirit of [Moro \(2012\)](#) to illustrate how sectoral shifts can result in lower output under high macro uncertainty relative to under low macro uncertainty. In the firm problem, we model a sectoral shift as a negative productivity shock in one sector but not in the other sector, and uncertainty is modeled via stochastic volatility. The household problem follows from [Moro \(2012\)](#).

IV.A *Households*

The economy is populated with a representative household that gains utility from consumption from two sectors of the economy. Household preferences are

$$U(C_{1,t}, C_{2,t}, N_t) = \log \left[\theta C_{1,t}^\xi + (1 - \theta)(C_{2,t} + \bar{Z})^\xi \right]^{\frac{1}{\xi}} + \psi \log(1 - N_t), \quad (5)$$

where $C_{j,t}$ denotes per-capita consumption in sector j , for $j \in \{1, 2\}$, in time t , N_t is labor hours in time t , $\xi < 1$, $\bar{Z} > 0$, and $\psi > 0$. The constant $\bar{Z}$ represents home production and θ is the consumption demand bias toward sector 1.

The household budget constraint is

$$P_{1,t}C_{1,t} + P_{2,t}C_{2,t} + P_{1,t}I_t = R_tK_t + W_tN_t, \quad (6)$$

where $P_{1,t}$ is the price in sector 1 in time t , $P_{2,t}$ is the price in sector 2 in time t , I_t is investment of the sector 1 good in time t , R_t is the rental price of capital in time t , K_t is capital in time t , and W_t is the wage rate in time t . The capital law of motion is

$$K_{t+1} = (1 - \delta)K_t + \frac{1}{\omega} \left(\frac{I_t}{K_t} \right)^\mu K_t, \quad (7)$$

where δ is the depreciation rate on capital, ω is the efficiency parameter, and μ is the capital

adjustment cost parameter.

The representative household, therefore, solves the following problem,

$$\max_{C_{1,t}, C_{2,t}, N_t, I_t} E_t \sum_{t=0}^{\infty} \beta^t U(C_{1,t}, C_{2,t}, N_t) \quad (8)$$

subject to the functional form of utility (equation (5)), the household budget constraint (equation (6)), and the capital law of motion (equation (7)). E_t is the expectations operator at time t and β is the subjective discount factor.

IV.B Firms

There are two sectors in the economy. Our exposition of each firm's problem follows from [Moro \(2012\)](#). Sector j 's production function for $j \in \{1, 2\}$ is

$$Y_{j,t} = B_{j,t} (K_{j,t}^\alpha N_{j,t}^{1-\alpha})^{\nu_j} \left(X_{j,t}^{\zeta_j} Z_{j,t}^{1-\zeta_j} \right)^{1-\nu_j}, \quad (9)$$

where $Y_{j,t}$ is output in sector j in time t , $K_{j,t}$ is capital in sector j in time t , $N_{j,t}$ is labor hours in sector j in time t , $X_{j,t}$ is sector j 's intermediate goods used by sector j in time t , $Z_{j,t}$ is sector i 's intermediate goods used by sector j in time t where $i \in \{1, 2\}$ and $i \neq j$, $0 < \alpha < 1$, $0 < \nu_j < 1$, $0 < \zeta_j < 1$, and $B_{j,t}$ is total factor productivity (TFP) in sector j in time t .

$$B_{j,t} = \bar{B}_j A_{j,t}, \quad (10)$$

where $\bar{B}_j$ is a constant $(1 + \gamma_j)$ in sector j and γ_j can be interpreted as productivity growth in sector j .¹⁹ $A_{j,t}$ is an AR(1) process in sector j ,

$$\log(A_{j,t}) = \rho_{A_j} \log(A_{j,t-1}) + \phi_t \epsilon_{j,t}, \quad (11)$$

¹⁹We note that the growth rates are isomorphic to long-run productivity levels. Throughout our analysis, we focus on the stationary version of the model.

where $0 < \rho_{A_j} < 1$ and $\epsilon_{j,t} \stackrel{i.i.d.}{\sim} N(0, \sigma_{A_j}^2)$. ϕ_t is an *economy-wide* stochastic volatility process that evolves according to

$$\log(\phi_t) = (1 - \rho_\phi) \log(\bar{\phi}) + \rho_\phi \log(\phi_{t-1}) + \epsilon_{\phi,t}, \quad (12)$$

where $0 < \rho_\phi < 1$, $\bar{\phi}$ is a constant, and $\epsilon_{\phi,t} \stackrel{i.i.d.}{\sim} N(0, \sigma_\phi^2)$.

A negative productivity shock in one sector and not in the other sector is our measure of a sectoral shift and the stochastic volatility component is our measure of economy-wide uncertainty in the model. Our choice of a sector-specific productivity shock to model a sectoral shift is motivated by our empirical analysis as well as the sectoral shift literature (Abraham and Katz, 1986; Loungani et al., 1990). The use of stochastic volatility to model uncertainty follows from the existing literature (Fernández-Villaverde et al., 2011; Caldara et al., 2012; Fernández-Villaverde et al., 2015). A negative productivity shock in sector 1 and not in sector 2 results in a reallocation of resources from sector 1 to sector 2 (i.e., a sectoral shift).²⁰ Additionally, an increase in the stochastic volatility component ϕ_t represents an increase in economy-wide uncertainty and it also widens the distribution of the realizations of $A_{j,t}$. In other words, the uncertainty shock amplifies/dampens the effect on productivity from a productivity shock. This is because the manufacturing productivity shock is multiplied by the economy-wide uncertainty variable.

Sector j 's representative firm's problem is as follows,

$$\begin{aligned} \max_{K_{j,t}, N_{j,t}, X_{j,t}, Z_{j,t}} \quad & [P_{j,t}Y_{j,t} - R_tK_{j,t} - W_tN_{j,t} - P_{j,t}X_{j,t} - P_{i,t}Z_{j,t}] \\ \text{s.t.} \quad & Y_{j,t} = B_{j,t} (K_{j,t}^\alpha N_{j,t}^{1-\alpha})^{\nu_j} (X_{j,t}^{\zeta_j} Z_{j,t}^{1-\zeta_j})^{1-\nu_j} \end{aligned} \quad (13)$$

where $P_{j,t}$ is the price of output produced in sector j at time t , R_t is the rental price of capital at time t , W_t is the wage rate at time t , and $P_{i,t}$ is the price of output produced in sector i at time t , for $i, j \in \{1, 2\}$ and $i \neq j$.

²⁰This is consistent with the sectoral shift literature (e.g., Lilien (1982) and Abraham and Katz (1986)). We thank an anonymous referee for bringing this to our attention.

IV.C Equilibrium and Market Clearing

Equilibrium conditions require that the following markets for capital, labor, and goods are cleared:

$$\int_0^1 K_t dh = K_t = K_{1,t} + K_{2,t},$$

$$\int_0^1 N_t dh = N_t = N_{1,t} + N_{2,t},$$

$$\int_0^1 C_{1,t} dh = C_{1,t},$$

$$\int_0^1 C_{2,t} dh = C_{2,t},$$

$$Y_{1,t} = C_{1,t} + I_t + X_{1,t} + Z_{2,t},$$

and

$$Y_{2,t} = C_{2,t} + X_{2,t} + Z_{1,t}.$$

IV.D Solution Method and Parameterization

The model is solved numerically by taking a third-order approximation around the steady state. Table 7 presents a list of parameters used in the model, which, with the exception of ρ_ϕ , draws on values used in [Moro \(2012\)](#).²¹ Since most of our parameter values are from [Moro \(2012\)](#), we do not elaborate on these parameter choices. To facilitate a more straightforward interpretation of sectoral shifts, the model's parameterization is such that sector 1 corresponds to manufacturing, and sector 2 corresponds to services.

[Table 7 Here]

IV.E Model Implications

We next illustrate the extent to which sectoral shifts impact macroeconomic performance during periods of low and high uncertainty by generating responses of aggregate output to a negative

²¹The value for ρ_ϕ is taken from [Caldara et al. \(2012\)](#).

one-percent shock to productivity in one sector and not in the other sector during periods of low and high uncertainty. For example, a negative productivity shock to sector 1 (manufacturing) but not to sector 2 (services) results in a reallocation of resources from sector 1 to sector 2 (i.e., a sectoral shift) and vice versa.

To generate the aggregate output response under the high uncertainty regime, we initiate a one-period negative one-percent shock to sector-specific productivity and a one-period positive ten-percent shock to the economy-wide stochastic volatility component (ϕ_t). We then calculate the percentage difference between this series and the steady state. To generate the aggregate output response under the low uncertainty regime, we initiate a one-period negative one-percent shock to sector-specific productivity and a one-period negative ten-percent shock to the economy-wide stochastic volatility component (ϕ_t). We then calculate the percentage difference between this series and the steady state.

[Figure 6 Here]

Figure 6 plots the responses of aggregate output to a negative one-percent shock to manufacturing (sector 1) productivity under the low and high uncertainty regimes.²² In response to the shock, the manufacturing sector (sector 1) becomes less productive relative to the services sector (sector 2), which thus, can be interpreted as a sectoral shift shock. This also results in a reduction in capital and investment. Since the combined reductions in manufacturing productivity and investment are not perfectly offset by a relative increase in productivity in the services sector, aggregate output falls. Although a sectoral shift results in a decrease in aggregate output under both the low and high uncertainty regimes, the aggregate output response is more negative during periods of high uncertainty (solid line) than during periods of low uncertainty (dashed line).²³ This is because the economy-wide uncertainty shock amplifies the effect on manufacturing productivity from the negative manufacturing productivity shock under high

²²Similar to [Moro \(2012\)](#), our measure of aggregate output in the model is real value-added aggregate output.

²³Figure B.4 in Appendix B plots the responses of aggregate output to a negative one-percent shock to services (sector 2) productivity under the low and high uncertainty regimes. In this case, the services sector (sector 2) becomes less productive relative to the manufacturing sector (sector 1), which can also be interpreted as a sectoral shift shock.

uncertainty relative to under low uncertainty. Overall, we find that the model prediction in Figure 6 is qualitatively consistent with the threshold regression and the local projections analyses in Section III.²⁴

[Figure 7 Here]

We next plot the responses of capital, investment, and labor hours to a negative one-percent shock to manufacturing productivity (i.e., sectoral shift) under the two uncertainty regimes in Figure 7.²⁵ In response to the shock, capital, investment, and labor hours all decline. These responses are amplified during periods of high uncertainty, which thus, results in a larger decline in output during periods of high uncertainty relative to during periods of low uncertainty.

[Figure 8 Here]

To further understand the mechanism in the model, we next plot, in Figure 8, the responses of sectoral output (manufacturing and services) and aggregate output to a negative one-percent shock to manufacturing productivity under the low and high uncertainty regimes.²⁶ The procedure used to generate these responses is the same as above. The responses under the low uncertainty regime are plotted on the left-hand side of Figure 8 and the responses under the high uncertainty regime are plotted on the right-hand side of Figure 8.

Two key insights stand out. First, in response to the negative manufacturing productivity shock, the manufacturing sector becomes less productive relative to the services sector regardless of the uncertainty regime. Because the manufacturing sector is now less productive, this results in a decline in investment, and therefore capital, which leads to a less efficient reallocation of

²⁴We also consider alternative specifications of the model where (1) there is no home production ($\bar{Z} = 0$) and (2) no capital adjustment cost ($K_{t+1} = I_t + (1 - \delta)K_t$). Overall, we find our baseline results to be qualitatively consistent with these changes regardless of whether the sectoral shift is from manufacturing to services (Figure B.5 in Appendix B) or vice versa (Figure B.6 in Appendix B).

²⁵Similar to [Moro \(2012\)](#), our measure of investment in the model is real value-added investment.

²⁶We follow from [Moro \(2012\)](#) and our measure of aggregate output is real value-added output and our measures of sectoral output are sectoral real value-added output net of investment. Figure B.7 in Appendix B plots the responses of sectoral output and aggregate output to a negative one-percent shock to services productivity under the low and high uncertainty regimes.

resources across the two sectors.²⁷ The initial increase in services output gradually decreases over time because goods in the services sector are complementary to manufacturing goods ($\xi = -1.5$) and also because final services production requires intermediate goods in both sectors as inputs.²⁸ Combined, these effects imply an overall decrease in aggregate output as observed in Figure 6. Second, moving from the low uncertainty regime to the high uncertainty regime amplifies the negative effects of a sectoral shift on manufacturing output and aggregate output. Even though services output is higher under the high uncertainty regime relative to under the low uncertainty regime on impact, this difference is not sufficiently large to make up for the reduction in manufacturing output. The combined effect is a more pronounced decrease in aggregate output when uncertainty is high relative to when it is low.

We next investigate whether the responses of aggregate output to sectoral shifts under the low and high uncertainty regimes depend on the *ex-ante* state of the economy. Specifically, we ask if the aggregate output responses to sectoral shifts under the low and high uncertainty regimes are different when aggregate output is *ex-ante* low (i.e., during recessionary periods) compared to when aggregate output is *ex-ante* high (i.e., during expansionary periods). This question is motivated by the empirical evidence that uncertainty is largely counter-cyclical (Schwert, 1990; Jurado et al., 2015). To evaluate this, we first simulate the model for 10,000 periods where productivity shocks from both sectors are present and use this simulated (base) series to identify the state of the economy. In particular, we classify these 10,000 periods into two states (low-output and high-output), each of which contains roughly 5,000 periods. The first state is a low-output state that contains periods where aggregate output is below or equal to the steady-state aggregate output value. The second state is a high-output state that contains periods where aggregate output is above the steady-state aggregate output value.

²⁷Investment, and therefore capital, decline more in response to the negative manufacturing productivity shock when there is no capital adjustment cost relative to when there is a capital adjustment cost because it is less costly to adjust investment and capital. Here, we would like to note that I_t is investment of the sector 1 (manufacturing) good. As a result, aggregate output declines more when there is no capital adjustment cost (Figure B.5 in Appendix B) relative to when there is a capital adjustment cost (Figure 6). The effect on aggregate output is amplified under high uncertainty relative to under low uncertainty because in addition to the other channels discussed, investment and capital decline more under high uncertainty relative to under low uncertainty. This is because in part, as noted above, the manufacturing productivity shock is multiplied by the economy-wide uncertainty variable.

²⁸The value of $\xi = -1.5$ is consistent with evidence from Rogerson (2008) and Duarte and Restuccia (2010).

From every point in each of these two states, we initiate a negative one-percent shock to manufacturing productivity. This shock can be interpreted as a sectoral shift that happens when *ex-ante* aggregate output is less than or equal to steady-state aggregate output (i.e., low-output state) or when *ex-ante* aggregate output is greater than steady-state aggregate output (i.e., high-output state). The procedure used to generate the responses under the low and high uncertainty regimes in each of these two states is the same as in Figure 6. The aggregate output responses to a sectoral shift are then averaged within each of the two output states to ensure that the plotted responses are not dependent on any particular draw of shocks. The average responses of aggregate output to a sectoral shift under the low and high uncertainty regimes for the two *ex-ante* aggregate output states (i.e., low-output vs. high-output) are plotted in Figure 9.²⁹

[Figure 9 Here]

One key insight from Figure 9 is that the aggregate output responses to a sectoral shift under the low and high uncertainty regimes are distinctively different from each other regardless of whether the shock is initiated when aggregate output is *ex-ante* low or *ex-ante* high. This prediction of the model is qualitatively consistent with our earlier empirical result that the asymmetric effects of sectoral shifts under the low and high uncertainty regimes are not driven solely by recessions.

V CONCLUSION

Using U.S. data, this paper finds that sectoral shifts, as proxied by increases in industry stock market dispersion, induce more depressed economic activity under high macro uncertainty relative to under low macro uncertainty, and these effects are statistically different from one another. Moreover, despite the counter-cyclical nature of uncertainty, we find that our results are not driven solely by recessions.

²⁹Figure B.8 in Appendix B plots the state-dependent average responses of aggregate output to a negative one-percent shock to services productivity under the low and high uncertainty regimes.

A two-sector dynamic stochastic general equilibrium model with sectoral shifts and stochastic volatility is able to qualitatively generate the asymmetric effects observed in the empirical analysis. Intuitively, a decrease in productivity in one sector, when not perfectly offset by a relative increase in productivity in another sector, will result in an overall decrease in aggregate output. This decline in aggregate output is predicted to be larger under high uncertainty relative to under low uncertainty. Consistent with the empirical evidence, the model also predicts that this result is not dependent on the *ex-ante* state of the economy; that is, whether the economy is in a low- or a high-output state *ex-ante*.

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Table 1: Industries Used in the Construction of Industry Stock Market Dispersion

Number	Industry	$\bar{c}_i$
1	Oil and Gas Producers	0.056
2	Oil Equipment, Services and Distribution	0.042
3	Chemicals	0.028
4	Forestry and Paper	0.016
5	Industries Metals and Mines	0.012
6	Mining	0.022
7	Construction and Materials	0.021
8	Aerospace and Defence	0.042
9	Electronic and Electrical Equipment	0.021
10	Industrial Engineering	0.028
11	Industrial Transportation	0.005
12	Support Services	0.061
13	Auto and Parts	0.166
14	Beverages	0.006
15	Food Producers	0.034
16	Household Goods, Home Construction	0.017
17	Leisure Goods	0.033
18	Personal Goods	0.002
19	Health Care Equipment and Services	0.033
20	Pharmaceuticals and Biotechnology	0.005
21	Food and Drug Retailers	0.020
22	General Retailers	0.001
23	Media	0.028
24	Travel and Leisure	0.016
25	Telecommunications	0.020
26	Electricity	0.016
27	Gas, Water and Multi-Utilities	0.030
28	Nonlife Insurance	0.013
29	Real Estate Investment, Services	0.046
30	REITs	0.031
31	Financial Services	0.044
32	Software and Computer Services	0.021
33	Technology Hardware & Equipment	0.065

Note: $\bar{c}_i$ denotes the average share of employees of industry i over time.

Table 2: Summary Statistics

	Mean	Median	St Dev	Min	Max	Obs
Industry Stock Market Dispersion ($\times 10^{-3}$)	0.08	0.07	0.04	0.02	0.31	408
Jurado et al. (2015) Macro (H-1) Uncertainty	0.64	0.63	0.08	0.55	1.07	408
Jurado et al. (2015) Macro (H-3) Uncertainty	0.78	0.77	0.08	0.68	1.21	408
Jurado et al. (2015) Macro (H-12) Uncertainty	0.91	0.90	0.05	0.85	1.15	408
Baker et al. (2016) EPU	108.26	102.21	31.43	57.20	245.13	408
VXO	20.10	18.35	8.27	7.87	61.41	396
Industrial Production Growth (%)	0.00	0.00	0.00	-0.01	0.01	408
Unemployment Rate Change (p.p. change)	-0.01	-0.03	0.08	-0.13	0.33	408
Federal Funds Rate (%)	3.63	3.75	2.81	0.07	9.85	408

Note: The time period is 1985M1-2018M12 for industry stock market dispersion, [Jurado et al. \(2015\)](#) macro (H-1, H-3, and H-12) uncertainty, [Baker et al. \(2016\)](#) EPU, industrial production growth, unemployment rate change, and the federal funds rate, and 1986M1-2018M12 for VXO.

Table 3: Industrial Production Growth Threshold Regressions with Macro (H-1, H-3, or H-12) Uncertainty as the Threshold Variable

Threshold Variable:	Macro Uncertainty								
	H-1 (1)	H-3 (2)	H-12 (3)	H-1 (4)	H-3 (5)	H-12 (6)	H-1 (7)	H-3 (8)	H-12 (9)
IP Growth (1-lag)	0.929*** (14.166)	0.932*** (14.480)	0.948*** (14.258)	0.900*** (13.998)	0.905*** (14.151)	0.915*** (14.302)	0.889*** (14.242)	0.891*** (14.214)	0.909*** (14.498)
IP Growth (2-lag)	-0.011 (-0.171)	-0.016 (-0.241)	-0.021 (-0.314)	0.023 (0.362)	0.016 (0.249)	0.006 (0.099)	0.032 (0.528)	0.027 (0.441)	0.013 (0.205)
Δ Rate (1-lag)	0.000 (1.179)	0.000 (1.280)	0.000 (1.347)	0.000 (0.423)	0.000 (0.309)	0.000 (0.636)	0.000 (1.133)	0.000 (0.844)	0.000 (1.153)
Δ Rate (2-lag)	0.000 (0.867)	0.000 (0.880)	0.000 (1.431)	0.000 (0.280)	0.000 (0.314)	0.000 (0.548)	0.000 (0.604)	0.000 (0.577)	0.000 (0.708)
Recession				-0.001*** (-3.655)	-0.001*** (-3.489)	-0.001*** (-3.699)			
Great Recession							-0.001*** (-3.391)	-0.001*** (-3.220)	-0.001*** (-3.238)
Regime 1: Low Uncertainty									
Dispersion (1-lag)	0.986 (1.067)	0.878 (0.930)	1.069 (1.164)	1.082 (1.166)	1.258 (1.355)	1.200 (1.301)	0.982 (1.062)	1.437 (1.550)	1.163 (1.233)
Dispersion (2-lag)	-0.271 (-0.311)	-0.321 (-0.384)	-0.498 (-0.593)	-0.278 (-0.321)	-0.352 (-0.420)	-0.196 (-0.232)	-0.411 (-0.474)	-0.328 (-0.387)	-0.190 (-0.219)
Constant	0.000 (1.578)	0.000* (1.726)	0.000 (1.399)	0.000 (1.615)	0.000 (1.631)	0.000 (1.519)	0.000 (1.611)	0.000 (1.355)	0.000 (1.308)
Regime 2: High Uncertainty									
Dispersion (1-lag)	-8.308** (-2.272)	-7.205* (-1.891)	-6.990** (-1.985)	-6.836** (-1.999)	-6.173** (-2.020)	-5.185 (-1.575)	-6.864* (-1.943)	-6.049** (-2.000)	-4.824 (-1.573)
Dispersion (2-lag)	-8.129*** (-2.821)	-9.250*** (-2.947)	-7.284* (-1.822)	-6.596** (-2.507)	-6.494** (-2.199)	-6.008* (-1.677)	-7.391*** (-2.727)	-6.382** (-2.309)	-5.334* (-1.881)
Constant	0.002*** (2.610)	0.001*** (2.604)	0.001** (2.381)	0.002*** (2.908)	0.002*** (2.802)	0.001** (2.479)	0.002*** (2.924)	0.001*** (2.686)	0.001** (2.568)
Observations	405	405	405	405	405	405	405	405	405
$\hat{\gamma}$	0.705	0.844	0.950	0.705	0.842	0.950	0.705	0.840	0.946
χ^2	12.585	12.878	8.922	9.700	10.602	7.212	10.138	11.627	8.478
p-value	0.000	0.000	0.003	0.002	0.001	0.007	0.001	0.001	0.004

Note: This table reports the threshold regression results from equation (2) with and without an NBER recession (or Great Recession) dummy variable, $k = 2$, the dependent variable is industrial production growth, and robust standard errors are used. The label Δ Rate represents the change in the nominal interest rate and the label Dispersion represents industry stock market dispersion. t statistics are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The maximal time period is 1985M1-2018M12 in specifications (1) through (9). χ^2 test statistics and their associated p-values are reported where the test hypotheses are $H_0: \sum_{j=1}^2 \delta_j = \sum_{j=1}^2 \theta_j$ and $H_A: \sum_{j=1}^2 \delta_j \neq \sum_{j=1}^2 \theta_j$.

Table 4: Industrial Production Growth Threshold Regressions with EPU or VXO as the Threshold Variable

Threshold Variable:	EPU (1)	VXO (2)	EPU (3)	VXO (4)	EPU (5)	VXO (6)
IP Growth (1-lag)	1.011*** (14.384)	0.998*** (13.905)	0.954*** (14.491)	0.936*** (14.383)	0.945*** (14.734)	0.923*** (14.365)
IP Growth (2-lag)	-0.068 (-0.941)	-0.050 (-0.687)	-0.028 (-0.417)	-0.005 (-0.070)	-0.024 (-0.349)	0.001 (0.021)
Δ Rate (1-lag)	0.000 (0.791)	0.000 (0.976)	-0.000 (-0.027)	0.000 (0.132)	0.000 (0.648)	0.000 (0.875)
Δ Rate (2-lag)	0.000 (1.122)	0.000 (1.261)	0.000 (0.149)	0.000 (0.414)	0.000 (0.569)	0.000 (0.742)
Recession			-0.001*** (-4.224)	-0.001*** (-4.601)		
Great Recession					-0.001*** (-3.647)	-0.001*** (-3.870)
Regime 1: Low Uncertainty						
Dispersion (1-lag)	0.743 (0.676)	0.319 (0.312)	1.018 (0.757)	1.548 (1.070)	0.579 (0.537)	0.441 (0.414)
Dispersion (2-lag)	-0.749 (-0.648)	-0.949 (-1.020)	-1.505 (-0.929)	-0.597 (-0.360)	-0.219 (-0.182)	-0.558 (-0.534)
Constant	0.000* (1.851)	0.000 (1.569)	0.000*** (2.829)	0.000 (0.546)	0.000** (2.261)	0.000* (1.746)
Regime 2: High Uncertainty						
Dispersion (1-lag)	-1.403 (-0.933)	-2.914 (-1.006)	-0.127 (-0.108)	-1.449 (-1.132)	-0.277 (-0.192)	-2.266 (-0.980)
Dispersion (2-lag)	-2.540** (-2.057)	-5.975* (-1.839)	-0.987 (-0.993)	-2.247** (-2.029)	-2.042* (-1.783)	-5.248** (-2.492)
Constant	0.000** (2.275)	0.001** (2.118)	0.000* (1.951)	0.001*** (3.591)	0.000** (2.011)	0.001*** (3.048)
Observations	405	394	405	394	405	394
$\hat{\gamma}$	99.136	28.330	88.044	18.740	99.136	26.940
χ^2	3.575	4.703	0.098	3.355	1.761	5.759
p-value	0.059	0.030	0.754	0.067	0.184	0.016

Note: This table reports the threshold regression results from equation (2) with and without an NBER recession (or Great Recession) dummy variable, $k = 2$, the dependent variable is industrial production growth, and robust standard errors are used. The label Δ Rate represents the change in the nominal interest rate and the label Dispersion represents industry stock market dispersion. t statistics are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The maximal time period is 1985M1-2018M12 in specifications (1), (3), and (5) and 1986M1-2018M12 in specifications (2), (4), and (6). χ^2 test statistics and their associated p-values are reported where the test hypotheses are $H_0: \sum_{j=1}^2 \delta_j = \sum_{j=1}^2 \theta_j$ and $H_A: \sum_{j=1}^2 \delta_j \neq \sum_{j=1}^2 \theta_j$.

Table 5: Unemployment Rate Change Threshold Regressions with Macro (H-1, H-3, or H-12) Uncertainty as the Threshold Variable

Threshold Variable	Macro Uncertainty								
	H-1 (1)	H-3 (2)	H-12 (3)	H-1 (4)	H-3 (5)	H-12 (6)	H-1 (7)	H-3 (8)	H-12 (9)
Δ UR (1-lag)	0.816*** (13.894)	0.821*** (14.076)	0.827*** (13.466)	0.800*** (13.926)	0.807*** (13.927)	0.817*** (13.410)	0.809*** (14.007)	0.811*** (14.018)	0.802*** (13.709)
Δ UR (2-lag)	0.106* (1.900)	0.100* (1.808)	0.104* (1.799)	0.124** (2.269)	0.116** (2.101)	0.114** (1.989)	0.115** (2.098)	0.111** (2.011)	0.121** (2.171)
Δ Rate (1-lag)	-0.009 (-1.636)	-0.010* (-1.821)	-0.008 (-1.427)	-0.006 (-1.011)	-0.007 (-1.201)	-0.005 (-0.903)	-0.009 (-1.563)	-0.010* (-1.675)	-0.007 (-1.244)
Δ Rate (2-lag)	-0.008 (-1.512)	-0.008 (-1.445)	-0.011** (-2.031)	-0.006 (-1.050)	-0.006 (-1.063)	-0.006 (-1.047)	-0.007 (-1.252)	-0.007 (-1.334)	-0.009* (-1.743)
Recession				0.014*** (3.707)	0.012*** (3.033)	0.019*** (4.751)			
Great Recession							0.015** (2.514)	0.012** (2.039)	0.016*** (2.683)
Regime 1: Low Uncertainty									
Dispersion (1-lag)	31.823* (1.704)	30.670* (1.647)	5.339 (0.262)	28.783 (1.547)	28.130 (1.512)	75.447 (0.781)	29.127 (1.556)	30.714* (1.648)	6.269 (0.312)
Dispersion (2-lag)	-30.579 (-1.521)	-28.646 (-1.401)	-21.797 (-1.000)	-34.077* (-1.716)	-30.215 (-1.490)	213.484** (2.184)	-31.794 (-1.550)	-27.568 (-1.350)	-18.726 (-0.860)
Constant	-0.003 (-1.209)	-0.003 (-1.272)	-0.002 (-0.736)	-0.002 (-1.173)	-0.003 (-1.299)	-0.030*** (-3.281)	-0.002 (-1.099)	-0.003 (-1.309)	-0.002 (-0.986)
Regime 2: High Uncertainty									
Dispersion (1-lag)	115.540 (1.607)	103.023 (1.452)	112.325** (2.390)	84.328 (1.214)	78.650 (1.164)	30.711 (1.577)	99.964* (1.659)	79.842 (1.287)	95.765** (2.372)
Dispersion (2-lag)	194.236*** (3.087)	189.298*** (2.800)	57.450 (0.969)	168.203*** (2.903)	155.313** (2.339)	-18.637 (-0.906)	154.684*** (2.686)	170.122*** (2.589)	38.762 (0.702)
Constant	-0.023** (-2.216)	-0.019** (-1.960)	-0.010 (-1.435)	-0.024** (-2.315)	-0.019** (-1.979)	-0.003 (-1.250)	-0.023** (-2.231)	-0.020** (-2.036)	-0.010 (-1.433)
Observations	405	405	405	405	405	405	405	405	405
$\hat{\gamma}$	0.708	0.844	0.936	0.708	0.844	0.863	0.705	0.844	0.936
χ^2	11.741	10.308	6.670	9.083	7.410	4.944	9.575	8.577	4.982
p-value	0.001	0.001	0.010	0.003	0.006	0.026	0.002	0.003	0.026

Note: This table reports the threshold regression results from equation (2) with and without an NBER recession (or Great Recession) dummy variable, $k = 2$, the dependent variable is the unemployment rate change (Δ UR), and robust standard errors are used. The label Δ Rate represents the change in the nominal interest rate and the label Dispersion represents industry stock market dispersion. t statistics are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The maximal time period is 1985M1-2018M12 in specifications (1) through (9). χ^2 test statistics and their associated p-values are reported where the test hypotheses are $H_0: \sum_{j=1}^2 \delta_j = \sum_{j=1}^2 \theta_j$ and $H_A: \sum_{j=1}^2 \delta_j \neq \sum_{j=1}^2 \theta_j$.

Table 6: Unemployment Rate Change Threshold Regressions with EPU or VXO as the Threshold Variable

Threshold Variable:	EPU (1)	VXO (2)	EPU (3)	VXO (4)	EPU (5)	VXO (6)
Δ Unemployment Rate (1-lag)	0.871*** (13.203)	0.870*** (13.564)	0.821*** (13.441)	0.815*** (13.040)	0.824*** (13.860)	0.819*** (13.517)
Δ Unemployment Rate (2-lag)	0.081 (1.306)	0.078 (1.288)	0.117** (2.034)	0.121** (2.063)	0.111* (1.953)	0.113* (1.950)
Δ Rate (1-lag)	-0.011* (-1.909)	-0.010 (-1.601)	-0.006 (-0.952)	-0.004 (-0.710)	-0.009 (-1.557)	-0.009 (-1.439)
Δ Rate (2-lag)	-0.012** (-2.104)	-0.011* (-1.709)	-0.008 (-1.346)	-0.004 (-0.714)	-0.010* (-1.742)	-0.006 (-1.080)
Recession			0.019*** (4.702)	0.020*** (5.025)		
Great Recession					0.021*** (3.549)	0.023*** (3.782)
Regime 1: Low Uncertainty						
Dispersion (1-lag)	-5.838 (-0.220)	9.340 (0.405)	-4.538 (-0.175)	78.743** (1.999)	0.253 (0.010)	85.588** (2.159)
Dispersion (2-lag)	11.671 (0.398)	5.358 (0.223)	5.186 (0.181)	68.957 (1.599)	3.752 (0.133)	46.290 (1.071)
Constant	-0.000 (-0.004)	-0.003 (-1.041)	-0.001 (-0.305)	-0.013*** (-3.220)	-0.001 (-0.269)	-0.012*** (-3.064)
Regime 2: High Uncertainty						
Dispersion (1-lag)	90.609*** (3.029)	122.278*** (3.029)	66.200** (2.359)	23.212 (0.991)	68.298** (2.522)	17.780 (0.789)
Dispersion (2-lag)	2.266 (0.075)	-20.922 (-0.373)	-14.105 (-0.506)	-22.916 (-0.969)	-1.891 (-0.067)	-14.610 (-0.592)
Constant	-0.010*** (-3.040)	-0.008 (-1.122)	-0.009*** (-2.701)	-0.002 (-0.715)	-0.009*** (-2.845)	-0.001 (-0.319)
Observations	405	394	405	394	405	394
$\hat{\gamma}$	104.305	28.180	104.305	17.290	104.305	17.440
χ^2	3.002	1.679	1.160	5.684	1.791	4.324
p-value	0.083	0.195	0.281	0.017	0.181	0.038

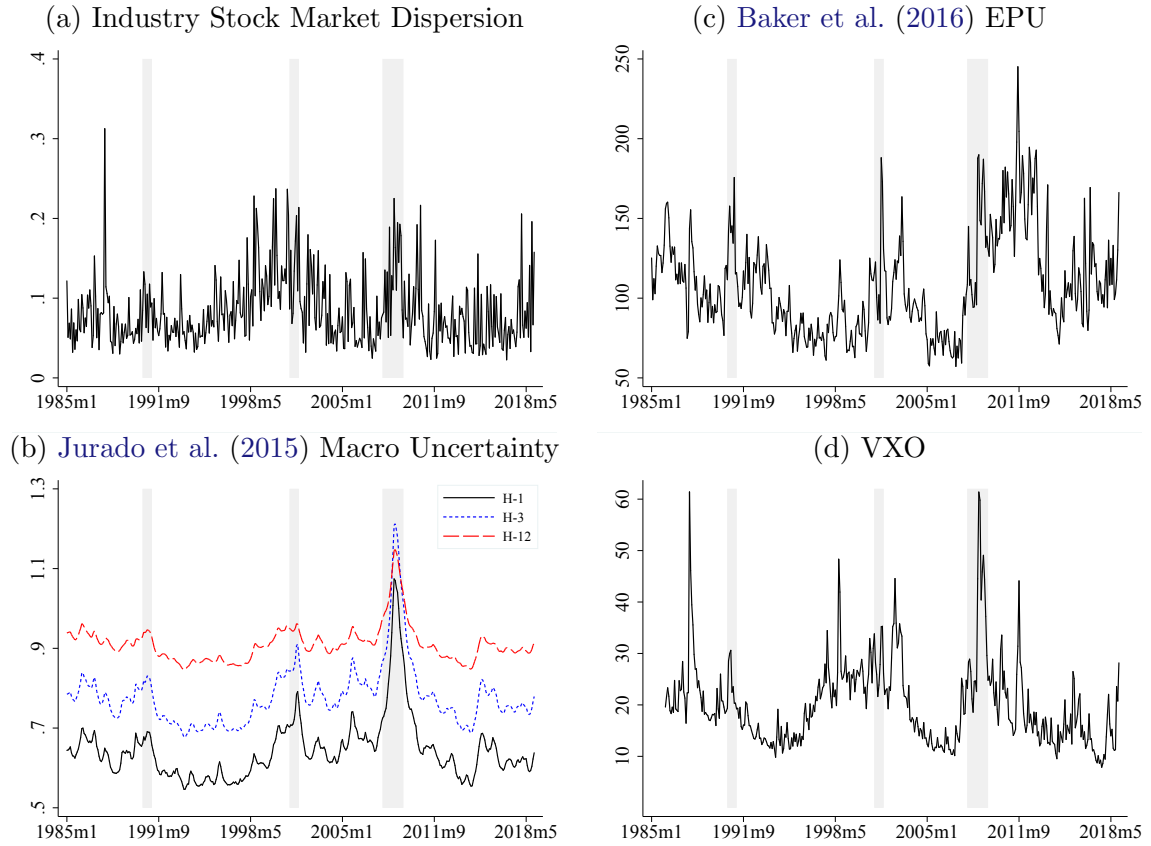
Note: This table reports the threshold regression results from equation (2) with and without an NBER recession (or Great Recession) dummy variable, $k = 2$, the dependent variable is the unemployment rate change, and robust standard errors are used. The label Δ Rate represents the change in the nominal interest rate and the label Dispersion represents industry stock market dispersion. t statistics are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The maximal time period is 1985M1-2018M12 in specifications (1), (3), and (5) and 1986M1-2018M12 in specifications (2), (4), and (6). χ^2 test statistics and their associated p-values are reported where the test hypotheses are $H_0: \sum_{j=1}^2 \delta_j = \sum_{j=1}^2 \theta_j$ and $H_A: \sum_{j=1}^2 \delta_j \neq \sum_{j=1}^2 \theta_j$.

Table 7: Parameter Values

Parameter	Description	Value
β	Subjective discount rate	0.985
ξ	Parameter governing the elasticity of substitution	-1.50
$\bar{Z}$	Home production of sector 2	1.273
ψ	Weight on leisure	0.159
θ	Weight on sector 1 in preferences	0.000023
δ	Depreciation rate	0.012
ω	Efficiency parameter in capital accumulation	2.026
μ	Capital adjustment cost parameter	0.80
α	Share of capital in value added	0.34
ν_1	Share of K_1 and N_1 in Y_1	0.40
ν_2	Share of K_2 and N_2 in Y_2	0.62
ζ_1	Share of X_1 in sector 1 intermediates	0.71
ζ_2	Share of X_2 in sector 2 intermediates	0.72
γ_1	Growth rate of sector 1 TFP	0.30
γ_2	Growth rate of sector 2 TFP	0.27
ρ_{A_1}	TFP autoregressive parameter in sector 1	0.95
ρ_{A_2}	TFP autoregressive parameter in sector 2	0.92
ρ_ϕ	Stochastic volatility auto-regressive parameter	0.90

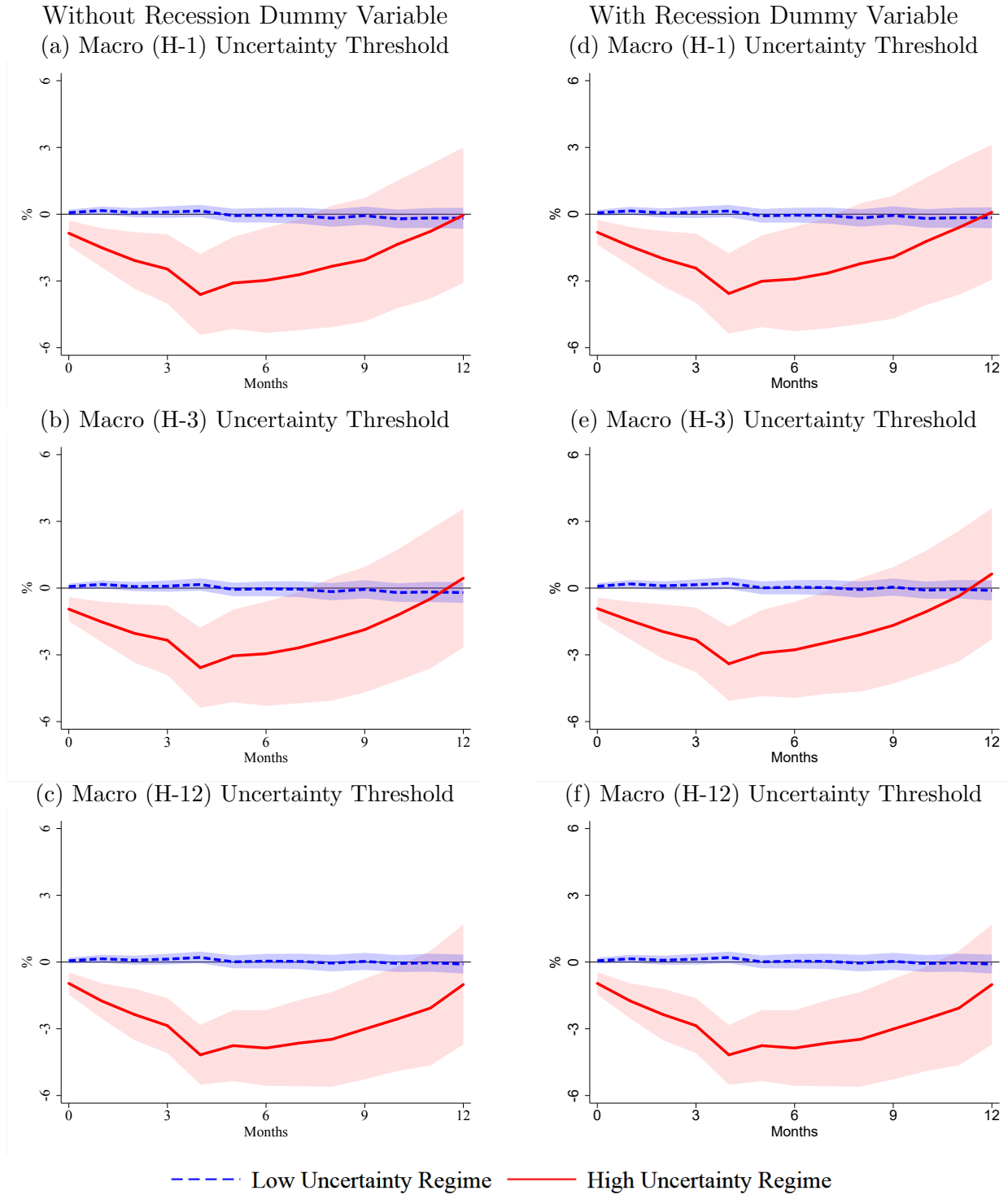
Note: With the exception of $\{\rho_\phi\}$, our parameter set draws on values used in [Moro \(2012\)](#).

Figure 1: Evolution of Selected Variables



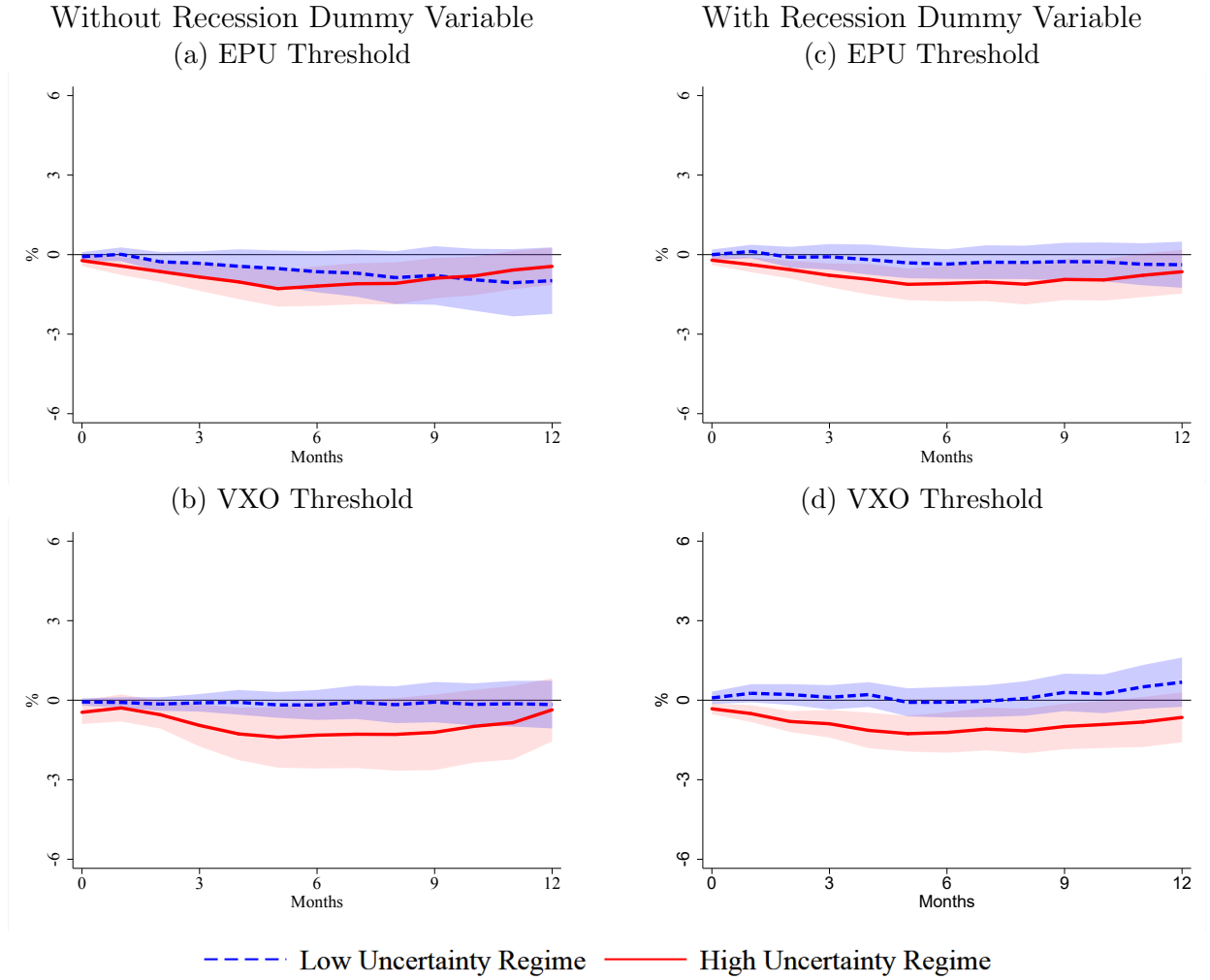
Note: The grey bars represent NBER recession dates. The time period is 1985M1-2018M12 for industry stock market dispersion, Jurado et al. (2015) macro uncertainty, and Baker et al. (2016) EPU, and 1986M1-2018M12 for VXO.

Figure 2: Responses of Industrial Production Growth to Industry Stock Market Dispersion Shock During Periods of Low and High Macro Uncertainty



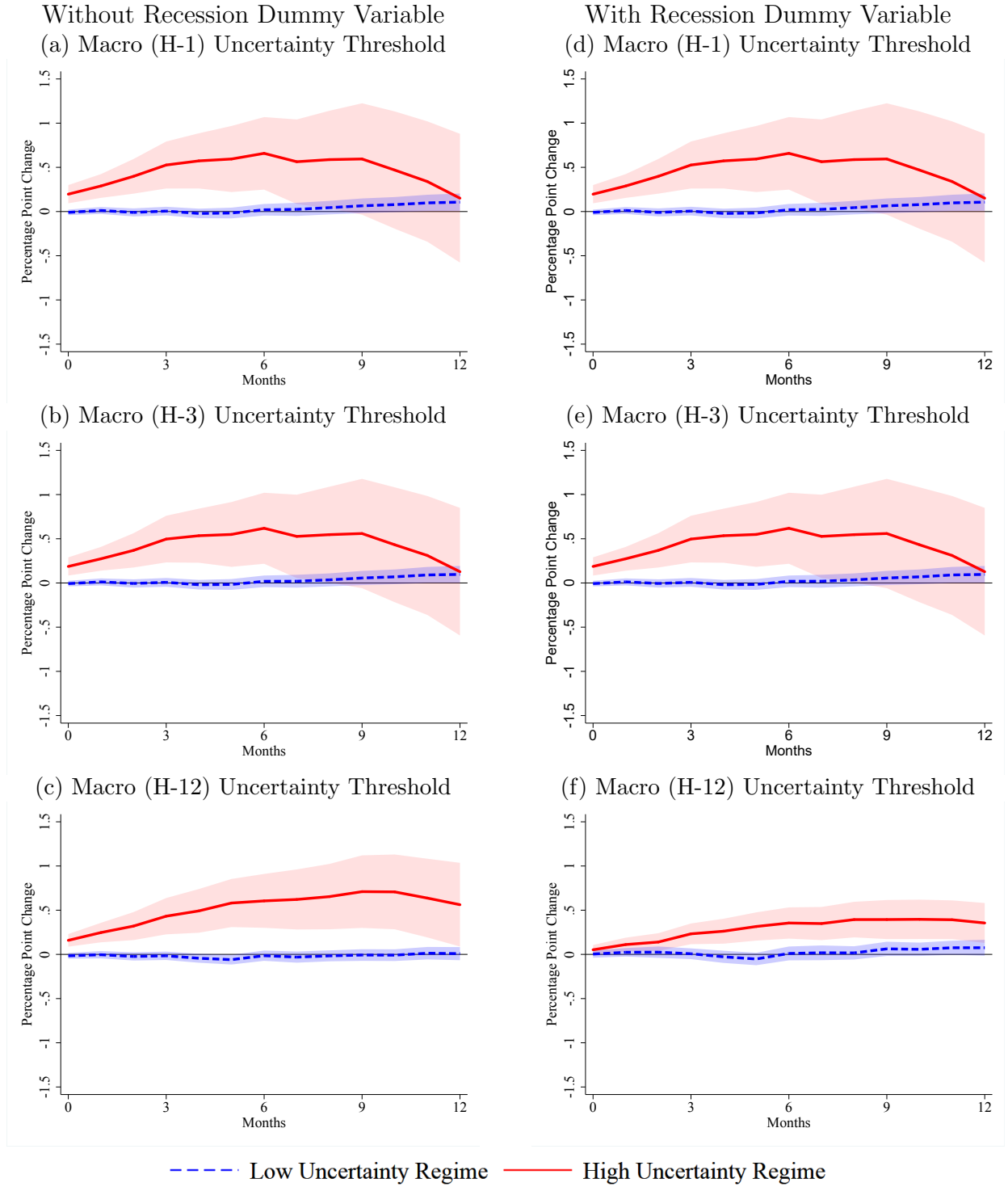
Note: This figure plots δ_h from equation (3) under low uncertainty (dashed line) and θ_h from equation (4) under high uncertainty (solid line) with and without an NBER recession dummy variable and $k = 2$. Shaded areas are 90% confidence bands. Robust standard errors are used. The maximal time period is 1985M1-2018M12.

Figure 3: Responses of Industrial Production Growth to Industry Stock Market Dispersion Shock During Periods of Low and High Market-Specific Uncertainty



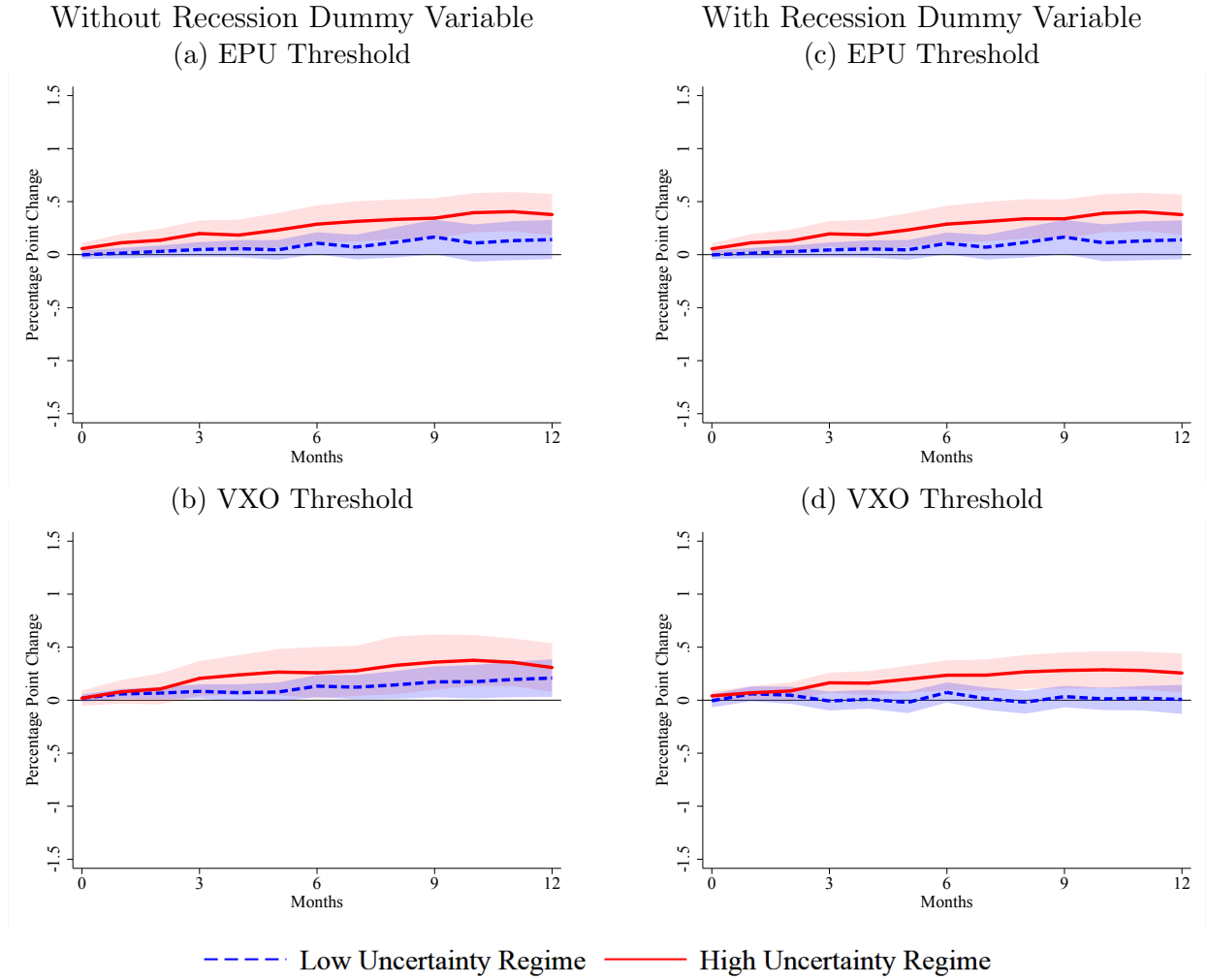
Note: This figure plots δ_h from equation (3) under low uncertainty (dashed line) and θ_h from equation (4) under high uncertainty (solid line) with and without an NBER recession dummy variable and $k = 2$. Shaded areas are 90% confidence bands. Robust standard errors are used. The maximal time period is 1986M1-2018M12 for VXO and 1985M1-2018M12 for EPU.

Figure 4: Responses of Unemployment Rate Change to Industry Stock Market Dispersion Shock During Periods of Low and High Macro Uncertainty



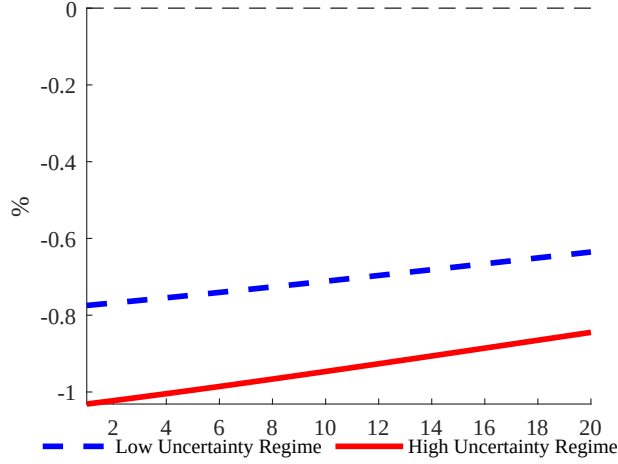
Note: This figure plots δ_h from equation (3) under low uncertainty (dashed line) and θ_h from equation (4) under high uncertainty (solid line) with and without an NBER recession dummy variable and $k = 2$. Shaded areas are 90% confidence bands. Robust standard errors are used. The maximal time period is 1985M1-2018M12.

Figure 5: Responses of Unemployment Change to Industry Stock Market Dispersion Shock During Periods of Low and High Market-Specific Uncertainty



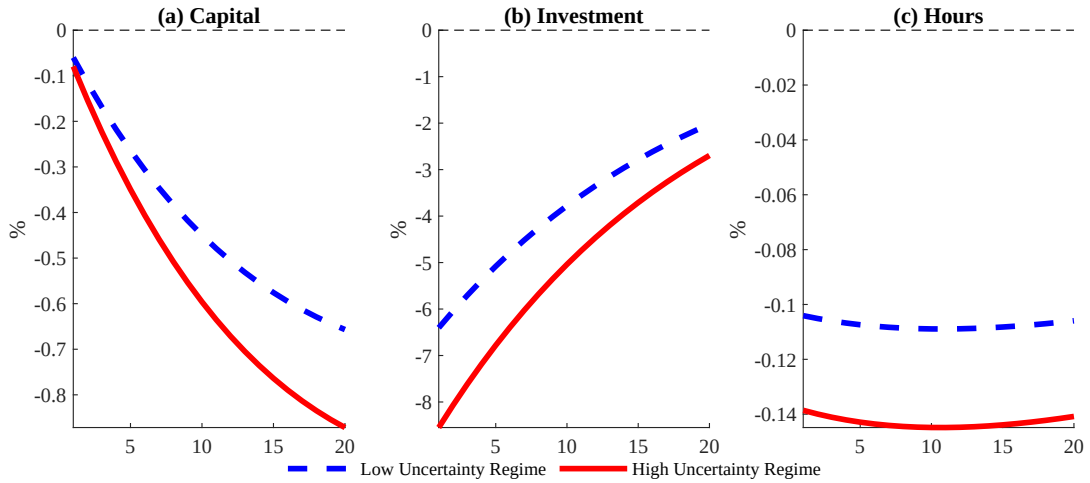
Note: This figure plots δ_h from equation (3) under low uncertainty (dashed line) and θ_h from equation (4) under high uncertainty (solid line) with and without an NBER recession dummy variable and $k = 2$. Shaded areas are 90% confidence bands. Robust standard errors are used. The maximal time period is 1986M1-2018M12 for VXO and 1985M1-2018M12 for EPU.

Figure 6: Responses of Aggregate Output to a Sectoral Shift (from a Negative Manufacturing Productivity Shock) During Periods of Low and High Uncertainty



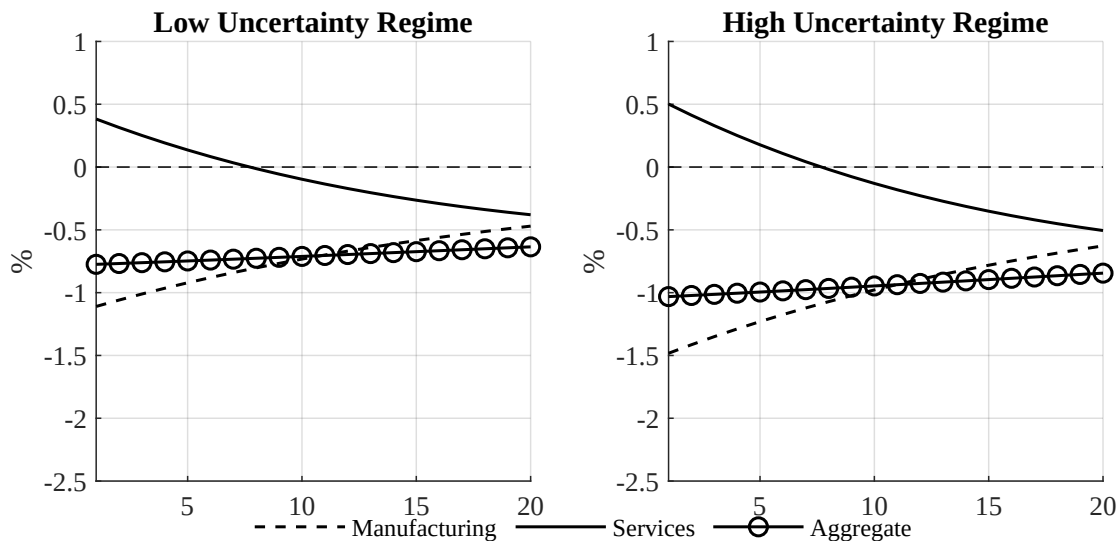
Note: This figure plots the responses of aggregate output to a negative one-percent shock to manufacturing productivity during periods of low and high uncertainty. To generate the aggregate output response under the high uncertainty regime, we initiate a one-period negative one-percent shock to manufacturing productivity and a one-period positive ten-percent shock to the economy-wide stochastic volatility component (ϕ_t). We then calculate the percentage difference between this series and the steady state. To generate the aggregate output response under the low uncertainty regime, we initiate a one-period negative one-percent shock to manufacturing productivity and a one-period negative ten-percent shock to the economy-wide stochastic volatility component (ϕ_t). We then calculate the percentage difference between this series and the steady state.

Figure 7: Responses of Capital, Investment, and Labor Hours to a Sectoral Shift (from a Negative Manufacturing Productivity Shock) During Periods of Low and High Uncertainty



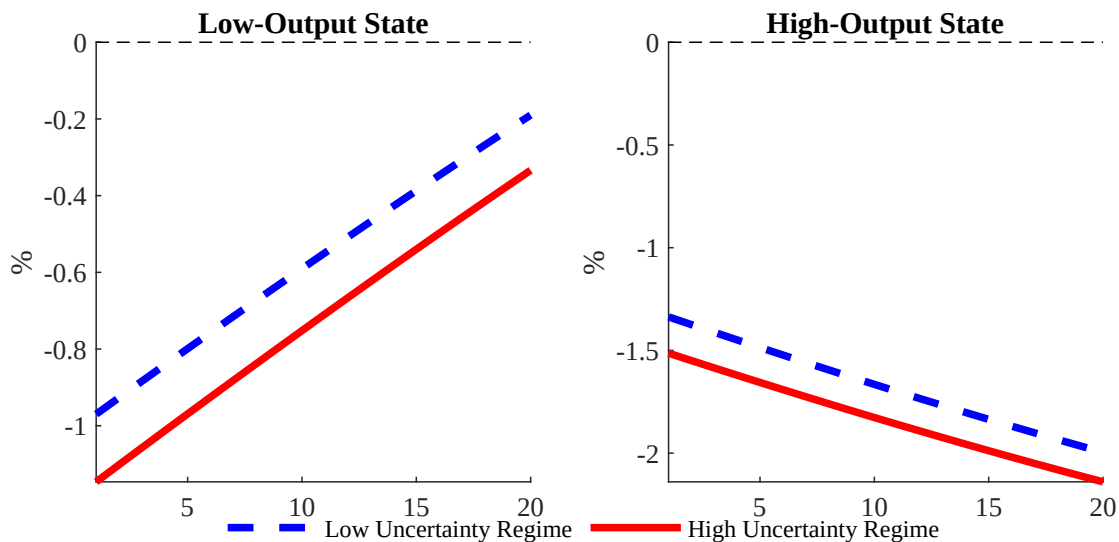
Note: This figure plots the responses of capital, investment, and labor hours to a negative one-percent shock to manufacturing productivity under the low and high uncertainty regimes. These responses are generated using the same procedure as in Figure 6.

Figure 8: Responses of Sectoral Output and Aggregate Output to a Sectoral Shift (from a Negative Manufacturing Productivity Shock) During Periods of Low and High Uncertainty



Note: This figure plots the responses of manufacturing output, services output, and aggregate output to a negative one-percent shock to manufacturing productivity under the low and high uncertainty regimes. These responses are generated using the same procedure as in Figure 6.

Figure 9: State-Dependent Average Responses of Aggregate Output to a Sectoral Shift (from a Negative Manufacturing Productivity Shock) During Periods of Low and High Uncertainty



Note: This figure plots the average responses of aggregate output to a negative one-percent shock to manufacturing productivity under the low and high uncertainty regimes when the *ex-ante* aggregate output is less than or equal to the steady-state aggregate output (i.e., low-output state) and when the *ex-ante* aggregate output is greater than the steady-state aggregate output (i.e., high-output state). The responses by the uncertainty regime are generated using the same procedure as in Figure 6.